DSA 210 Project Progress Report

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Project Title: The Correlation Between Domestic Player Representation in Top Clubs and National Team Success (1986-2024)

1. Motivation and Problem Definition

The central question of this project is whether a national team's success is dependent on the "Domesticity Index" of its local league. In an era where football is increasingly globalized, many assume that having more local players in top domestic clubs benefits the national team. However, with the rise of elite European leagues (Top 5), this relationship has become more complex. This project investigates if a higher domestic player ratio in clubs correlates with success in FIFA World Cups and UEFA European Championships.

2. Data Collection and Characteristics

Following the proposed methodology, a hybrid data collection approach was implemented:

- **National Performance:** Sourced from Kaggle’s "International Football Results" dataset.
- **Squad Demographics:** Extracted from Transfermarkt and official archives using a Python-based pipeline (Requests/BeautifulSoup).
- **Club Trophies:** Manually curated data on international club successes (UCL, UEL) was integrated to enrich the analysis.
- **Data Cleaning:** Python's **Pandas** library was used to merge these disparate sources, handle missing values (especially for historical market values), and standardize the "Domesticity Index" for 160 primary observations.

3. Methodology

The project follows a rigorous data science pipeline:

1. **Exploratory Data Analysis (EDA):** Visualizing the relationships between success and various factors (Age, Market Value, League Quality).
2. **Globalization Trend Analysis:** Monitoring the change in player distribution over decades.
3. **Hypothesis Testing:** Using Pearson Correlation and Independent T-Tests to verify the significance of the Domesticity Index.
4. **Machine Learning:** Implementing a **Random Forest Regressor** to identify the most critical features for predicting tournament success.

4. Analysis and Findings (Results)

4.1. Exploratory Data Analysis

- **FIFA Rank & Market Value:** Showed the strongest correlations ($r = -0.27$ and $r = 0.21$ respectively), proving that pre-tournament form and financial value are key indicators.
- **Top 5 League Ratio:** A positive correlation ($r = 0.19$) confirms that having players in elite leagues is a significant advantage.

4.2. Hypothesis Testing: The Domesticity Index

- **Initial Hypothesis:** Higher Domesticity Index = Higher Success.
- **Statistical Result:** The Pearson correlation was nearly zero ($r = -0.048$), and the T-test P-value was > 0.05.
- **Conclusion:** We fail to reject the null hypothesis. Domesticity Index does not have a statistically significant impact on national team success in the modern era.

4.3. Machine Learning & Feature Importance

The Random Forest model identified **FIFA Rank**, **Market Value**, and **Club Trophy Count** as the top three predictors.

5. Conclusion and Future Work

The project successfully demonstrated that modern football success is driven by globalization and financial strength rather than domestic player concentration. While the initial hypothesis regarding the Domesticity Index was not supported by the data, the discovery of the "Globalization Trend" (+12.63\% increase in Top 5 League Ratio) provides a more accurate explanation for contemporary success.

- **Limitations:** Historical financial data for the 1980s was estimated due to lack of digital archives.
- **Future Work:** Future models could include "Manager Experience" and "Player Fatigue Metrics" to improve predictive accuracy.